Universal perturbations on six classifiers

Analysis of the results: reproduction of Moosavi-Dezfooli et al. (2017), CNN / Transformer comparison, dominant classes and a Gaussian-noise defense. November 2026. (English translation of analyse_resultats.pdf.)

In short. (1) A single invisible perturbation (±10/255 per pixel) fools **72 to 83%** of the 40,000 ImageNet validation images for the six networks tested, against 6 to 18% for a control of the same norm. (2) The three attention-based networks (MaxViT, Swin, ViT) are **harder to fool** than the three CNNs (72–75% against 80–83%): more passes are needed, and their perturbation generalises less well. (3) Four very different architectures send the images to **the same class, "brain coral"**; ResNet-50 to "spider web", ViT to "scoreboard". (4) Gaussian noise σ = 20/255 at the input brings the fooling rate of ResNet-50 from 84.8% down to 10.5% for 4.5 points of accuracy, but only against an attacker who **ignores** the defense: the experiment with an adaptive attacker (EOT) was still running when this analysis was written.

1. Protocol

We follow the algorithm of the paper: we go through a set of images X; for each image that the current perturbation v does not fool yet, DeepFool computes the smallest step that changes its class, we add it to v, then project v onto the ℓ_∞ ball of radius ξ. Passes over X are repeated until 80% of X is fooled.

Parameter	Value
Set X (to compute v)	10,000 images: 10 per class of the ILSVRC2012 validation set
Evaluation	the 40,000 other validation images (disjoint from X)
Norm and radius	ℓ_∞, ξ = 10/255 (largest possible ℓ_2 norm: 15.2)
Stopping rule	fooling rate ≥ 80% on X (δ = 0.2)
DeepFool	10 candidate classes, overshoot 0.02, at most 10 iterations
Models	pretrained torchvision weights, unmodified; input 224 × 224
Hardware	school machine "espagne" (RTX A4000)

Fooling rate: fraction of the images whose predicted class changes when v is added. **Control**: the same values as v, but with the pixels shuffled at random; it has exactly the same norm and measures what the mere "amount" of perturbation does, without its structure.

2. Results on the six models

Model	Family	Clean acc.	Acc. with v	Fooling (eval.)	Fooling on X	Passes	ℓ_2	Control
ResNet-18	CNN	68.8%	18.0%	**79.6%**	82.1%	2	11.05	17.7%
ResNet-50	CNN	80.5%	17.1%	**81.0%**	82.3%	4	10.46	9.5%
ConvNeXt-B	modern CNN	83.9%	15.5%	**82.9%**	83.5%	1	10.30	7.1%
MaxViT-T	hybrid conv. + attention	83.5%	23.0%	**74.9%**	80.1%	3	9.07	5.9%
Swin-B	windowed Transformer	83.3%	23.9%	**73.8%**	80.5%	5	9.21	7.9%
ViT-B/16	Transformer	80.8%	25.7%	**71.9%**	72.8%*	9*	11.85	7.6%

* ViT never reached 80% on X: the fooling rate plateaued around 73% from the 6th pass on (figure 2) and the computation was stopped during the 10th pass. The perturbation evaluated is the one of the last checkpoint.

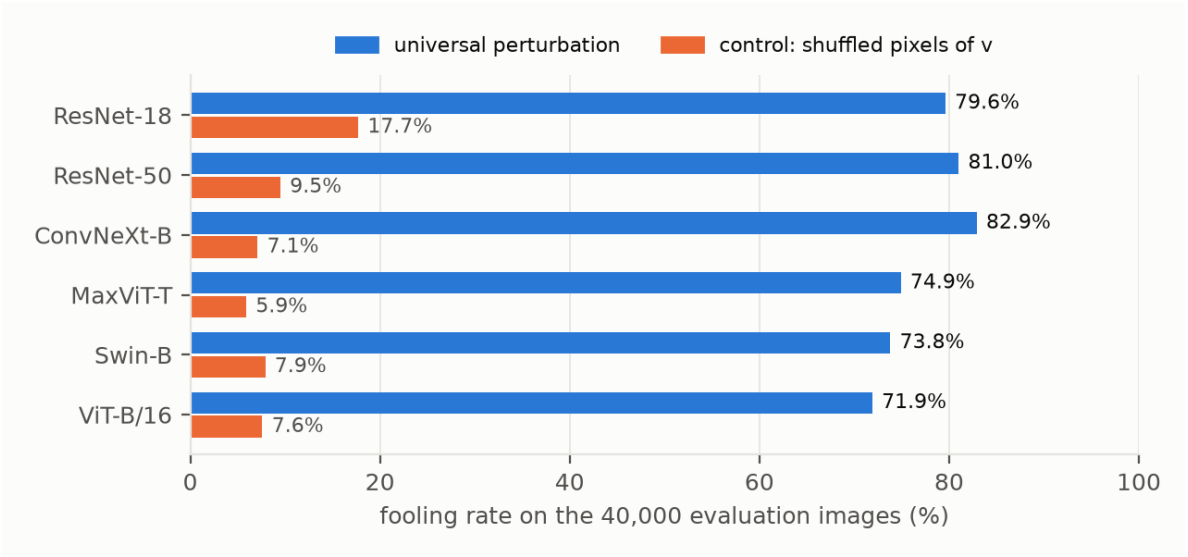


Figure 1. Fooling rate on the 40,000 evaluation images, compared with the control of the same norm.

2.1 The reproduction works on every architecture

For all six networks, the perturbation fools 4 to 12 times more than the control of the same norm: it is its **direction**, not its size, that fools. ResNet-50 (81.0%) is within the range of the paper (78–84% for VGG, GoogLeNet and ResNet-152). The control of ResNet-18 is higher (17.7%): this network, the least accurate, is also the most sensitive to any noise.

2.2 Attention-based networks resist better

The three CNNs reach 80–83%, the three attention-based networks 72–75%. The gap shows in two ways:

- **The computation is harder.** ConvNeXt exceeds 80% on X after the first pass, ResNet-18 after two. MaxViT needs three, Swin five, and ViT does not get there in nine (figure 2).
- **The perturbation generalises less well.** Between X and new images, the CNNs lose 0.6 to 2.5 points; MaxViT loses 5.2 and Swin 6.7. The perturbation “sticks” more to the images of X.

ConvNeXt is instructive: it is a CNN modernised to look like a Transformer (large kernels, modern training recipe), and it is the most accurate model of the set. Yet it is the **easiest** to fool. Robustness therefore does not follow accuracy, and what distinguishes the three robust models is attention. This is a hypothesis: with one model per family and one run per model, the effect of the architecture cannot be separated from that of the training recipe.

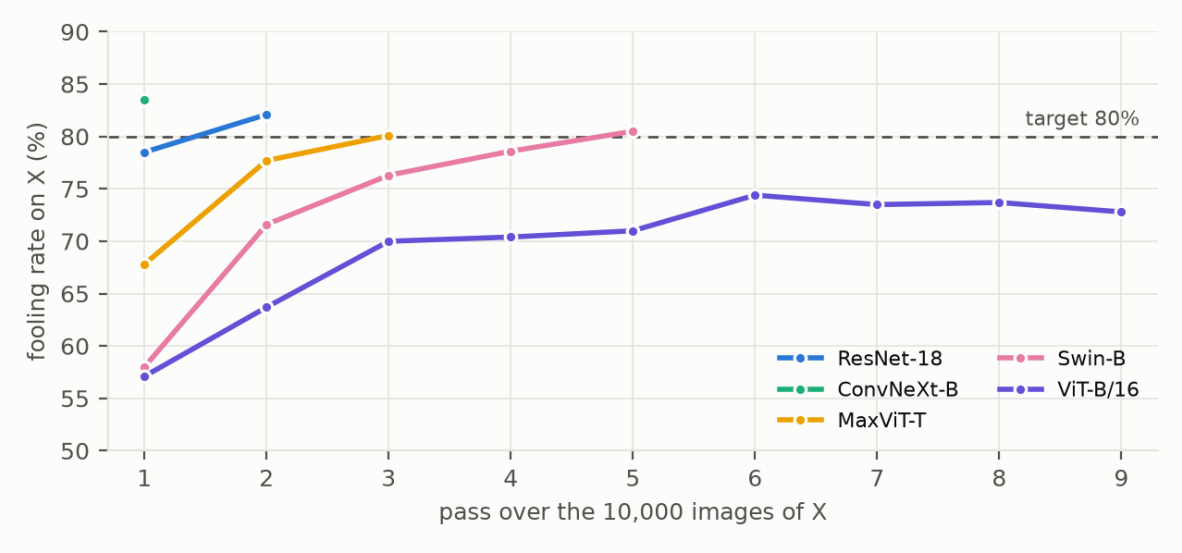


Figure 2. Fooling rate on X at the end of each pass. For MaxViT, the value of pass 1 is the one stored in the resume checkpoint.

A word of caution on the comparison. The algorithm stops as soon as 80% of X is fooled: it compares the models at a fixed target, not at a fixed effort. A CNN stopped after one pass might have exceeded 85% with five passes. A fairer comparison would fix the number of passes (option `--max_passes`) or use the gradient-based algorithm, which does not stop on an image that is already fooled.

3. Dominant classes

Model	Dominant class	Share of the fooled images	Other classes
ResNet-18	brain coral	18.4%	shower curtain 15.7%, mask 5.3%, peacock 5.1%
ResNet-50	spider web	97.5%	—
ConvNeXt-B	brain coral	97.6%	spider web 0.2%
MaxViT-T	brain coral	98.9%	—
Swin-B	brain coral	97.8%	—
ViT-B/16	scoreboard	96.8%	prayer rug 0.2%

As in the paper, the perturbation does not push each image towards a neighbouring class: it sends almost everything to **one single class**. The striking fact is that **four architectures out of six** (a small CNN, a modern CNN, a hybrid, a windowed Transformer) choose the same one: brain coral. These perturbations were computed independently. This suggests that the dominant class depends more on the **training data** than on the architecture: in ImageNet, “brain coral” may be a class defined almost only by a texture (tight, winding ridges) that a pattern repeated over the whole image is enough to imitate. “Spider web” is of the same kind (thin threads on a uniform background). For ResNet-50 we checked it: on a uniform gray image, v alone gives “spider web” with probability 0.99.

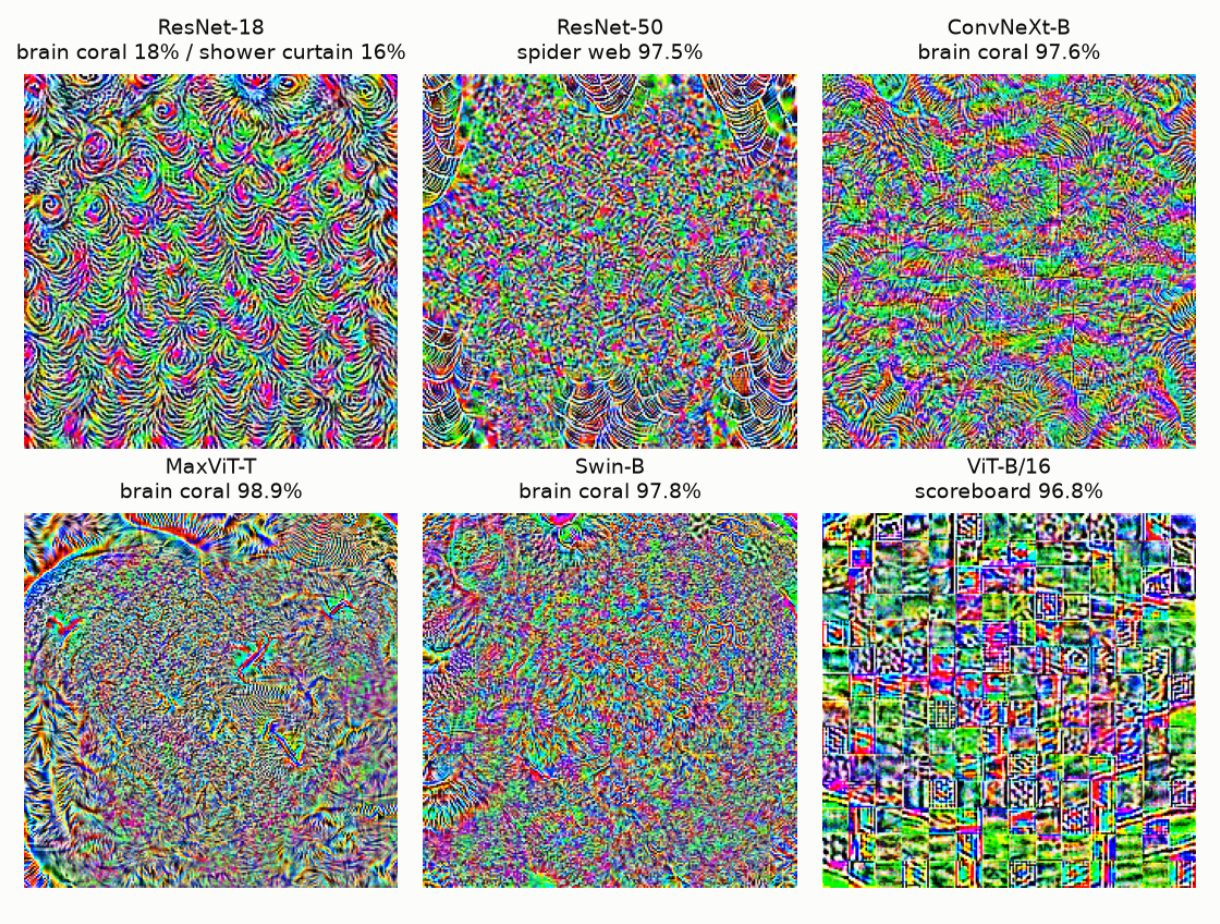


Figure 3. The six perturbations, stretched to [0, 1] to be visible (real values within ±10/255).

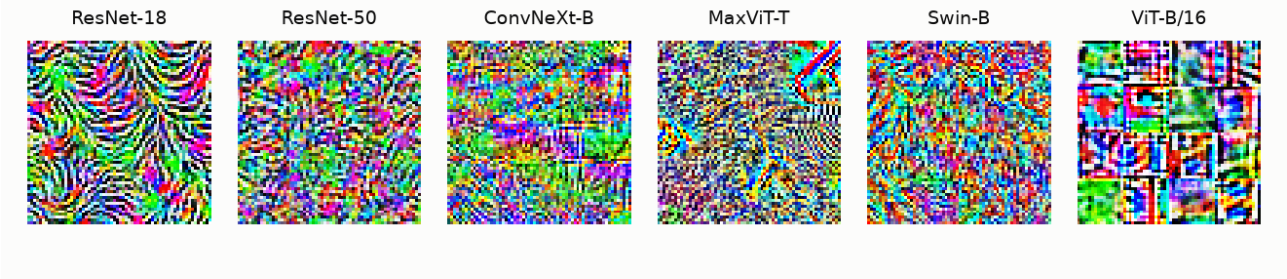


Figure 4. Zoom on 64 × 64 pixels at the centre of each perturbation.

The images support this reading. The perturbation of ResNet-18 draws **wavy ridges**, very close to a brain coral, and this pattern appears again, finer and noisier, in ConvNeXt, MaxViT and Swin. ResNet-50 shows **arcs of threads** on the edges, like a web. ViT stands apart: its perturbation is a **grid of 14 × 14 squares** of 16 pixels, exactly the grid of patches the model cuts the image into. Each square contains its own pattern and the edges are sharp. The class “scoreboard” (a board made of rectangular cells) is a plausible interpretation of this grid. ResNet-18, the only one without a clear dominant class, is also the least accurate: its perturbation hesitates between several textures (coral, shower curtain, peacock feathers).

These interpretations remain visual hypotheses. Two simple tests would check them: the gray-image test (`class_stats.py`) for each model, and the **transfer matrix**: does the perturbation of Swin also send ConvNeXt to “brain coral”?

4. Gaussian-noise defense (ResNet-50)

The perturbation of the paper is built from **minimal** DeepFool steps: each image is pushed just beyond the boundary (overshoot 0.02). It should therefore be fragile. To test this, an independent Gaussian noise of standard deviation σ is added to each image before the classifier. We use the ImageNet perturbation of ResNet-50 and the Imagenette validation set (10 classes, 1,000 images: 100 per class), without recomputing v: the attacker **ignores** the defense.

σ (/255)	A: clean + noise, correctly classified	B: attacked + noise, correctly classified	C: fooling rate under the noise	E: classified “spider web”
0	85.1%	15.1%	84.8%	83.1%
2	87.5%	16.6%	82.7%	81.4%
5	87.1%	24.1%	75.0%	72.5%
10	84.2%	53.7%	42.5%	37.7%
20	**80.6%**	**77.0%**	**10.5%**	**0.1%**
40	57.0%	56.4%	26.4%	0.0%

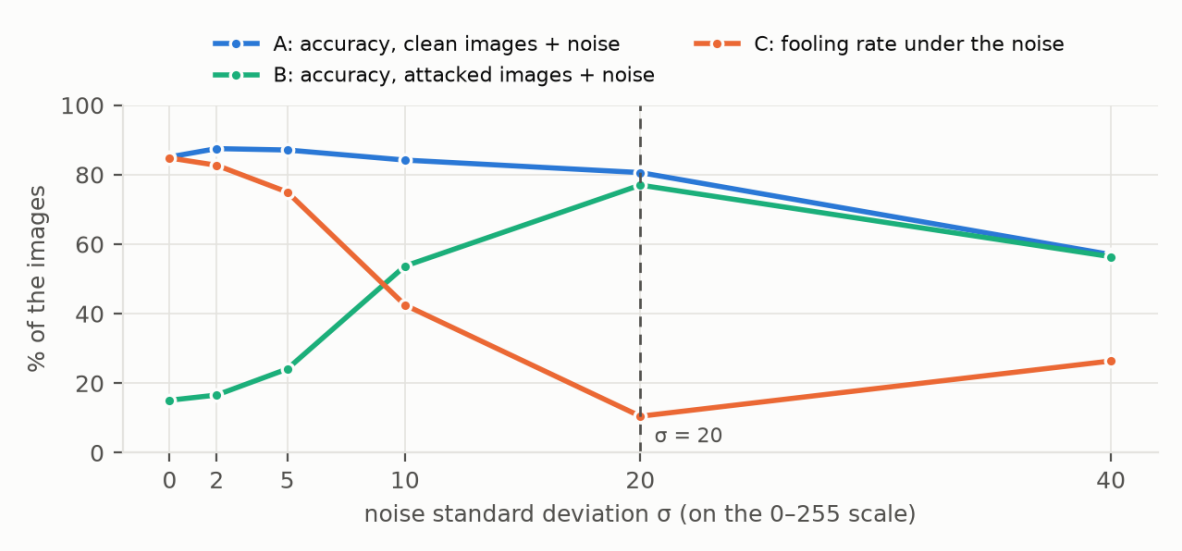


Figure 5. Cost (A) and effectiveness (B, C) of the defense depending on the standard deviation of the noise.

The noise neutralises the perturbation. At σ = 20/255, the fooling rate drops from 84.8% to 10.5%, the level of the control (9.5%), and no image is classified “spider web” any more. The accuracy on clean images only loses 4.5 points. This is consistent with the margin argument: a minimal step does not survive a noise larger than itself. Beyond that, the noise itself becomes harmful: at σ = 40, the clean accuracy drops to 57% and C rises again (26.4%), because the noise alone already changes many predictions.

This result does not prove that the defense is good. The literature (Athalye et al. 2018) has shown that random defenses of this kind almost always fall against an attacker who knows them and computes the perturbation on average over the noise (Expectation Over Transformation, EOT). For a universal perturbation, Xu & Singh (ICML 2024) show that a perturbation robust to transformations exists. The only question that matters is therefore: **what is left of the defense against a perturbation computed with EOT?**

5. Experiment in progress: adaptive attacker (EOT)

The code was extended (`--noise_std`, `--eot_samples`, `--algo sgd`). On 500 Imagenette images, with ResNet-50 and a defense at σ = 20, three perturbations are computed on the same images, then evaluated under the noise:

Perturbation	Construction	Question
Reference	algorithm of the paper, without noise (at most 10 passes)	point of comparison on 500 images
DeepFool + EOT	each DeepFool step is the average over 4 noisy copies (at most 5 passes)	is the algorithm of the paper enough, once adapted?
Gradient + EOT	gradient ascent on the cross-entropy, averaged over 4 noise draws	does a perturbation with margin break the defense?

To watch in particular: the fooling rate under the noise (column C at σ = 20), and the image of the robust perturbation. If the defense falls, which class dominates, and does the perturbation become a coarser (low-frequency) texture that survives the noise?

Update (translator's note). This experiment has since been run, first on Imagenette (inconclusive: the setting was too small) and then on ImageNet, where the adaptive attacker breaks the defense (45–50% fooling under σ = 20 at ξ = 10/255, 86% at ξ = 16/255). See the project overview, section 6.

6. Limitations and next steps

Limitations. One run per model: on a GPU, two identical computations do not give exactly the same perturbation, and the random order of the images changes the result. Differences of one or two points are not significant. The models were stopped at different stages (target reached or plateau). X is drawn from the validation set and not from the training set as in the paper. The noise defense was only evaluated on ResNet-50 and Imagenette.

Next steps, by order of interest:

- **Result of the EOT experiment** (in progress): does the noise defense resist an adaptive attacker, and at what cost?
- **Dominant class under EOT**: how the dominant class evolves when the perturbation must survive the noise. Our literature review found no study on this point (moderate confidence).
- **Transfer matrix** between the six models, and the gray-image test for each: does the “brain coral” of Swin fool ConvNeXt?
- **Noise defense on the six models** (`eval_noise.py`): do Transformers, already more robust, need less noise?
- **Comparison at equal effort**: same number of passes for all models, or the gradient-based algorithm.